Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 165 (2019) 532–540

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

Design of New Multi-Column 5,5:4 Compressor Circuit Based on Double-Gate UTBSOI Transistors

Rekib Uddin Ahmed, Sheba Diamond Thabah, Prabir Saha*

National Institute of Technology Meghalaya, Shillong, Meghalaya 793003, India

Abstract

In the modern electronic devices, compressor circuits are used to speed up the partial product addition (PPA) stage in the multipliers. This paper presents an architecture of multi-column 5,5:4 compressor whose new transistor level implementation has been carried out through the ultra-thin-body silicon-on-insulator (UTBSOI) transistors. Such compressor has been simulated in Cadence-spectre, and the performance parameters like power consumption and delay has been calculated at the rising and falling edge transition of input pulses. Advantage of the UTBSOI-compressor is further clarified from its comparison with the CMOS based design (CMOS-compressor). Simulation results reveal that the PDP exhibited by the UTBSOI-compressor is reduced by 79.89% than that of CMOS counterparts. Furthermore, to showcase the advantage of such compressor, it has been applied in the multiplier for PPA which offers ≈11.5% improvement in terms of power-delay-product (PDP) in the Virtex 6 platform.

Keywords: BSIM-IMG; compressor; multi-column; transmission-gate; UTBSOI.

1. Introduction

Modern electronic devices rely on fast and efficient arithmetic circuits to execute dedicated operations on binary signals. With the trends on increasing mobility and performance in portable devices, speed improvement (lesser

* Corresponding author. Tel.: +919485177005; fax: +913642501113.
E-mail address: sahaprabir1@gmail.com

10.1016/j.procs.2020.01.027

delay) and power reduction have become a challenging task for researchers which have been achieved through scaling down the conventional MOSFETs. Continued scaling down of MOSFETs has now reached its limits due to the short-channel effects (SCE) [1]. Various advanced MOSFET structures like double-gate (DG), gate-all-around (GAA), FinFET, and UTBSOI transistors are proposed to overcome the limitations of SCEs [2]. The UTBSOI transistor is one of the dual-gated devices which offers better immunity towards SCEs and now is being used at production level by leading semiconductor industries [3]. Fig. 1 shows a conceptual schematic of the double-gate (DG)-CMOS [4] based inverter designed using the UTBSOI transistors. The back gate of the UTBSOI provides superior controllability of gates over the shorter channel region. The back gate also provides the flexibility to attain the multiple threshold voltage control, which was earlier achieved through different channel doping concentration in conventional MOSFETs [3]. To utilize the benefits of UTBSOI, a fast and accurate model for the device is required. Department of Electrical Engineering and Computer Science at the University of California, Berkley has developed the industry-standard model for UTBSOI known as the BSIM-Independent multi-gate (IMG) [5]. The BSIM-IMG can be successfully adopted in the simulators like spectre or spice in order to simulate any DG-CMOS based circuits designed using the UTBSOIs.

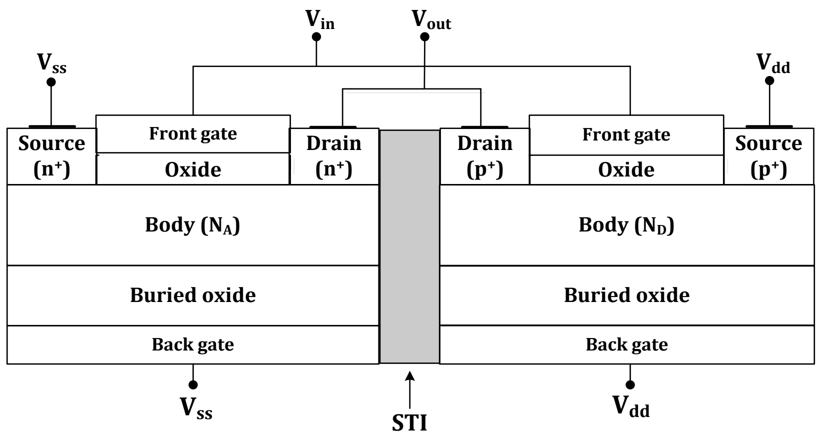


Fig. 1 Conceptual schematic showing the DG-CMOS based inverter designed using the UTBSOI

In microprocessors and digital signal processors, multipliers are used to perform the dedicated arithmetic operations like convolution, correlation, and filtering, etc. The binary multiplication is carried out in two stages, namely: 1) partial product generation, and 2) partial product addition (PPA). Conventionally, the PPA was accomplished through the use of half adders (HA) and full adders (FA) which caused most of the delay, chip area, and power consumption in the multipliers. In order to overcome these limitations, compressors are used in many multiplier circuits which counts the number of "ones" present in the input vector. Different types of compressor like 3:2, 4:3, and 5:3 have been reported in the literature [6-9] are used in 8×8 bit multiplier circuits. Addition to this, large-sized compressors like 7:3 and 15:4 are also proposed to design 16×16 and 32×32 bit multipliers [10-12]. Fig. 2 shows the different PPA methods used in an 8 × 8 multiplier based on the Wallace tree architecture. Considering the columns five and six [Fig. 2], where columns of five bits are to be added in a single stage. It can be accomplished by using two 5:3 compressors, and it will generate six output bits. Either it can be performed using the HAs and FAs which will produce eight output bits. If the same operation is carried out using the 5,5:4 compressor [13], then the number of generated output bits will be four. Thus the multi-column compressor like 5,5:4 is more useful in column reduction in the PPA stage. Moreover, the 5,5:4 compressor is useful in large-sized multipliers, squarer circuits, etc.

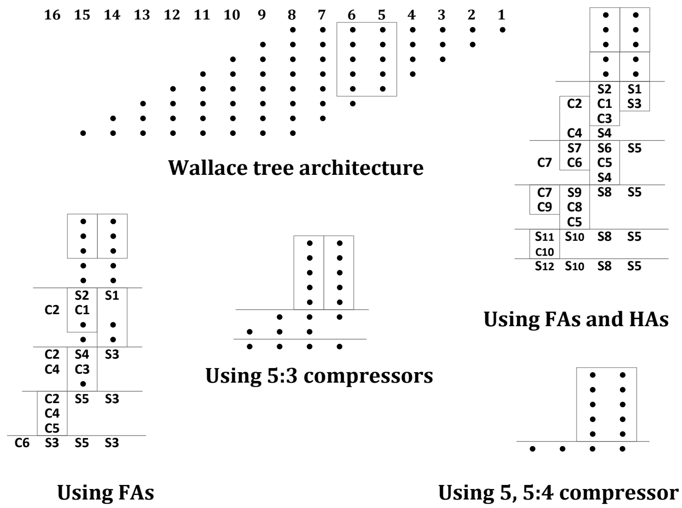


Fig. 2. Different methods of column reduction of PPA in the 8×8 Wallace tree multiplier.

In this paper, a new DG-CMOS based 5,5:4 compressor is designed using the UTBSOI transistors (UTBSOI-compressor). The BSIM-IMG model has been utilized to simulate the transistor level design of such a compressor. The delay and power consumption by the UTBSOI-compressor have been analyzed at the rising and falling transition of input pulses. Moreover, the functionality of the compressor is emphasized through performing the PPA of five bits [Fig. 2] using the device Virtex 6, simulated in Xilinx ISE 14.2 design environment through VHDL coding. Use of the 5,5:4 compressor in performing the PPA has resulted in 11.53% improvements in terms of power-delay product (PDP) over the other method using the FAs. Advantage of the UTBSOI-compressor is clarified from the comparison of its performance parameter with the CMOS based 5,5:4 compressor (CMOS-compressor). The PDP achieved by the UTBSOI-compressor is 79.89% lesser than that of CMOS-compressor.

2. Design of the 5,5:4 compressor

In the 5,5:4 compressor, two arrays of input vectors $[X_1, X_0] = [(a_1b_1c_1d_1e_1), (a_0b_0c_0d_0e_0)]$ can be fed, where each vector is of five bits as shown in Fig. 3. The number of "ones" present in each array are counted first and finally appended to generate the output bits q_3, q_2, q_1, and q_0. Inside the compressor, there exists intermediate signals f_{i2}, f_{i1}, and f_{i0}, which represent the number of "ones" present in the respective input arrays with i values ranging from 0

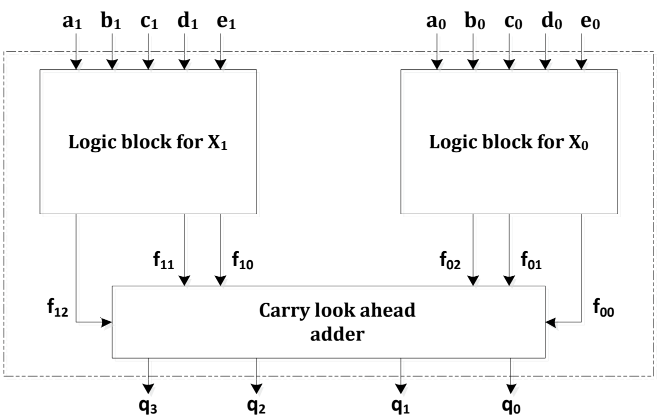


Fig. 3. Block diagram of 5, 5:4 compressor.

Table 1. Randomly selected set of combinations of $\boldsymbol{X_1}, \boldsymbol{X_0}$ and the expected result at outputs $[q_3 q_2 q_1 q_0]$.

Input vectors $X_1 X_0$	Combinations			
$a_1 a_0$	00	01	10	11
$b_1 b_0$	01	10	10	11
$c_1 c_0$	10	11	10	11
$d_1 d_0$	11	00	10	11
$e_1 e_0$	00	01	10	11
$\sum X_1 X_0 = q_3 q_2 q_1 q_0$	0110	0111	1010	1111

to 1. Table 1 shows a few possible combinations of the input vector and the corresponding output bits. The Boolean expressions for the intermediate signals are obtained as:

$$f_{i0} = a_i \oplus [(b_i \oplus c_i) \oplus (d_i \oplus e_i)] \tag{1}$$

$$f_{i1} = \bar{a}_i b_i d_i \bar{e}_i + \bar{a}_i \bar{b}_i c_i d_i + \bar{a}_i b_i c_i \bar{d}_i + a_i \bar{c}_i d_i \bar{e}_i + \bar{b}_i c_i \bar{d}_i e_i + \bar{a}_i \bar{c}_i d_i e_i + a_i b_i \bar{d}_i \bar{e}_i + a_i \bar{b}_i c_i \bar{e}_i + b_i \bar{c}_i \bar{d}_i e_i + a_i \bar{b}_i \bar{c}_i e_i \tag{2}$$

$$f_{i2} = [b_i c_i d_i e_i + a_i c_i d_i e_i + a_i b_i c_i e_i + a_i b_i d_i e_i + a_i b_i c_i d_i]\overline{f_{i1}} \tag{3}$$

The outputs of the presented compressor are obtained by adding the intermediate signals by carry-look ahead adder.

$$\begin{array}{cccc} f_{12} & f_{11} & f_{10} & 0 \\ 0 & f_{02} & f_{01} & f_{00} \\ \hline q_3 & q_2 & q_1 & q_0 \end{array}$$

Finally, Boolean expression for the output bits of the compressor q_3, q_2, q_1 and q_0 are expressed as:

$$q_0 = f_{00} \tag{4}$$

$$q_1 = (f_{01} \oplus f_{10}) \tag{5}$$

$$q_2 = (f_{02} \oplus f_{11}) \oplus f_{01} f_{10} \tag{6}$$

$$q_3 = f_{12} \oplus [f_{02} f_{11} + (f_{02} \oplus f_{11}) f_{01} f_{10}] \tag{7}$$

3. Results and Discussion

In order to emphasize the advantage of 5,5:4 compressor, PPA of two columns having five bits [Fig. 2] has been performed using HAs, FAs, 5:3 compressors, and the 5,5:4 compressor in Xilinx ISE 14.2 simulated through VHDL coding. For delay and power analyses, simulation is performed using the device Virtex 6 with power supply of 1V at 50º C. The total number of look-up tables (LUT) present in the Virtex 6 is 474240 [14] and its supply power obtained from simulation is 4.447 W. The power consumed while performing the PPA in Virtex 6 is calculated using the relation:

$$Power = \frac{Supply\ Power \times Number\ of\ LUTs\ used}{Total\ number\ of\ LUTs} \tag{8}$$

Table 2. Comparison of power consumption, delay, and PDP obtained from different methods of PPA in the device Virtex 6.

PPA methods	Components used	Supply Power (W)	Total number of LUTs	Number of LUTs used	Power (μW)	Delay (ns)	PDP (pJ)
Method 1	FAs	4.471	474240	9	84.39	2.773	0.234
Method 2	FAs and HAs	4.471	474240	12	112.52	3.215	0.361
Method 3	5:3 compressors	4.471	474240	12	112.52	2.249	0.253
Method 4	5, 5:4 compressor	4.471	474240	9	84.39	2.453	0.207

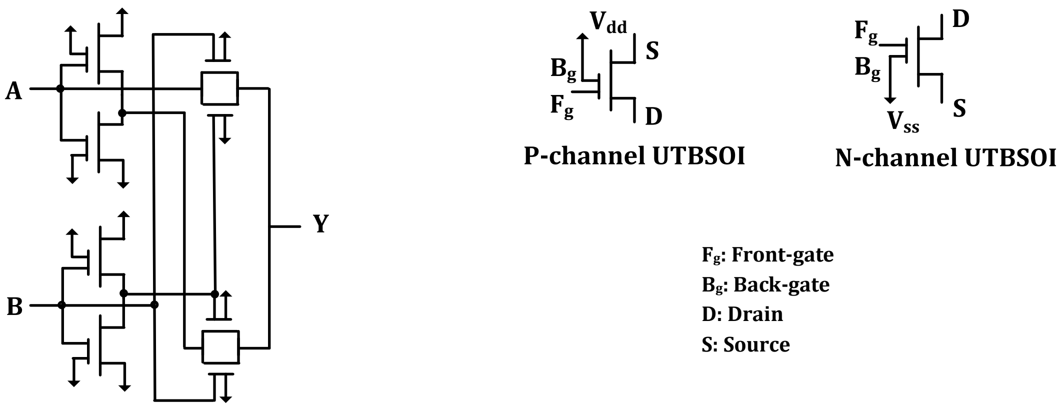


Fig. 4. Schematics showing the DG-CMOS based (UTBSOI) XOR gate.

The power consumption and delay analyses for different methods of PPA are given in Table 2, where it has been observed that the column reduction in PPA using the 5, 5:4 compressor (method 4) has the least PDP = 0.207 pJ, along with 11.53% improvement over the method using the FAs (method1).

The DG-CMOS based design of the compressor is simulated at the transistor level using the UTBSOIs by utilizing the BSIM-IMG model. At the preliminary stage, the Verilog-A code of BSIM-IMG is compiled in Cadence-spectre, and the generated symbol for UTBSOI is simulated accordingly to validate the outputs with the provided benchmark tests [5]. The UTBSOI-compressor is simulated under the power supply of 0.5V along with the input bits of amplitude 0.5V having a time period of 10 μs with rising and fall time of 10 ps. Device dimensions considered for the UTBSOI are: channel length $L = 55$ nm, channel width $W = 10$ μm, silicon body thickness t_{si} =12 nm, and thickness of oxide layer at the front gate (t_{oxf}) and back gate (t_{oxb}) interfaces are 2.4 nm and 10 nm respectively [5]. The XOR operation in Boolean expressions (1, 5−7) is performed using the transmission gate (TG) based XOR gates [15] as shown in Fig. 4. The TGs are chosen because they are meant for low power consumption [16] and fewer transistor counts [17]. The glitches at the outputs are reduced by increasing the load capacitances (C_L) with the penalty of increased delay and power consumption.

Table 3 presents the delay and power consumption observed at the rising and falling edge transition of the same set of combinations [Table1]. Fig. 5 shows the variation of average delay and power consumption calculated from Table 3 with respect to different C_L values. The delay and power consumption observed at rising edge transition are larger as compared to that observed at falling edge transition. The delay increases rapidly as the C_L increases, varies with a slope of 41.34 ns/pF. The variation of power consumption with respect to C_L is almost static, as can be seen from Fig. 5 (b). The device parameters of the UTBSOIs are considered the same as given in the benchmark test [5]. The simulation has been further carried out by making a few changes in device parameter values in order to reduce power consumption. The new set of device parameter values after making changes in [5] are given in Table 4. Table 5 shows the delay and power consumption exhibited by UTBSOI-compressor with the new set of device parameter values. New device parameter values of the UTBSOI have resulted in 99.37% and 97.75% reduction in

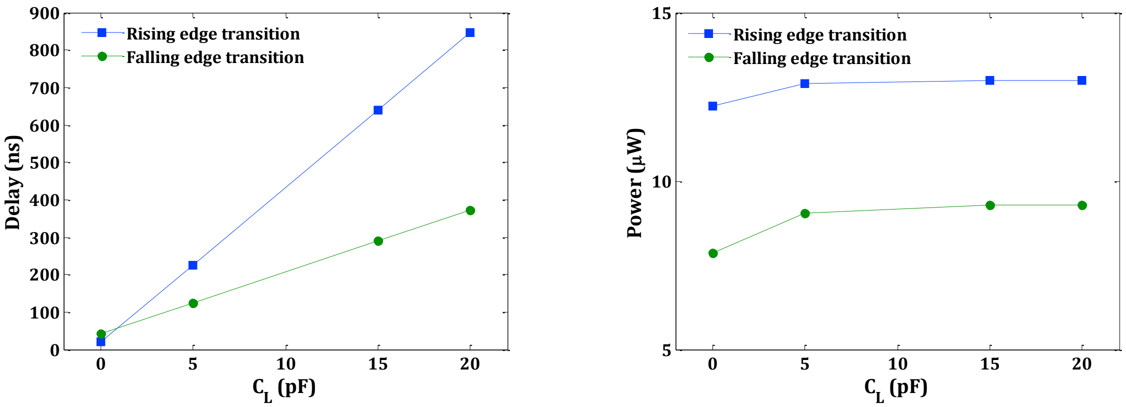
Table 3.Delay and power consumption exhibited by the UTBSOI-compressor with different values of C_L at rising and falling edge transition of input vector for the combinations given in Table 1.

Rising edge transition:

X_1					X_0					C_L							
										0 pF		5 pF		15 pF		20 pF	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)
0	0	1	1	0	0	1	0	1	0	25.9	13.44	252.9	13.32	709.2	13.46	937.7	13.37
0	1	1	0	0	1	0	1	0	1	24.9	11.5	251.7	11.41	708.2	11.57	937.0	11.51
1	1	1	1	1	0	0	0	0	0	11.7	8.819	153.1	9.198	437.4	9.179	579.9	9.211
1	1	1	1	1	1	1	1	1	1	17.8	15.19	244.0	17.64	700.4	17.8	928.8	17.81
				Average:						20.07	12.23	225.42	12.89	638.8	13.00	845.85	12.97

Falling edge transition:

X_1					X_0					C_L							
										0 pF		5 pF		15 pF		20 pF	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)	Delay (ns)	Power (μW)
0	0	1	1	0	0	1	0	1	0	68.3	9.805	158.5	9.44	339.7	9.436	430.5	9.465
0	1	1	0	0	1	0	1	0	1	65.1	4.013	155.3	5.068	336.4	4.738	427.4	4.513
1	1	1	1	1	0	0	0	0	0	18.8	6.141	75.2	5.976	190.0	5.977	247.2	5.974
1	1	1	1	1	1	1	1	1	1	19.7	11.54	110.4	15.73	291.6	17.06	382.4	17.21
				Average:						42.97	7.874	124.85	9.053	289.42	9.302	371.87	9.29



(a) (b)

Fig. 5. Variation of (a) average delay and (b) average power consumption of the UTBSOI- compressor with respect to different values of C_L .

Table 4. The new device parameter values of UTBSOI transistor (The rest parameter values are kept the same as given in [5]).

Parameters	UTBSOI (BSIM-IMG) n-channel	UTBSOI (BSIM-IMG) p-channel
Channel length (L)	55 nm	55 nm
Channel length (W)	165 nm	330 nm
Source diffusion area	45375 nm^2	90750 nm^2
Drain diffusion area	45375 nm^2	90750 nm^2
Source diffusion periphery	880 nm	1210 nm
Drain diffusion periphery	880 nm	1210 nm
Threshold voltage (V_{th})	0.329	−0.365

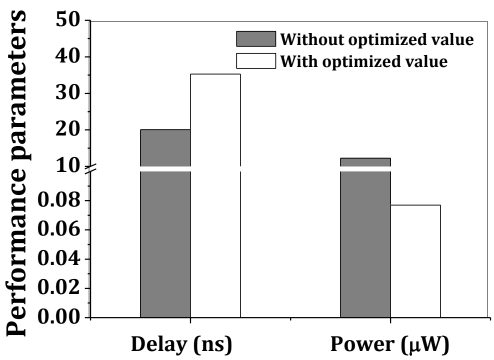
Table 5. Delay and power consumption exhibited by the UTBSOI-compressor (without C_L) at rising and falling edge transition of input vector for the combinations given in Table 1 with the optimized values of UTBSOI device parameters (after making changes in the device parameter values given in [5]).

Rising edge transition:

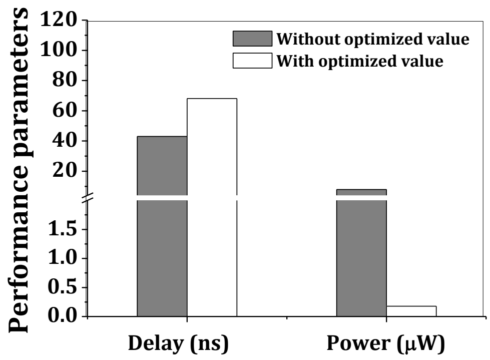
X_1					X_0					Without optimized value [Table 3]		With optimized value	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (μW)	Delay (ns)	Power (nW)
0	0	1	1	0	0	1	0	1	0	25.9	13.44	53.3	183.7
0	1	1	0	0	1	0	1	0	1	24.9	11.5	50.9	37.95
1	1	1	1	1	0	0	0	0	0	11.7	8.819	14.8	29.4
1	1	1	1	1	1	1	1	1	1	17.8	15.19	22.2	56.65
Average:										20.07	12.23	35.3	76.93

Falling edge transition:

X_1					X_0					Without optimized value [Table 3]		With optimized value	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (μW)	Delay (ns)	Power (nW)
0	0	1	1	0	0	1	0	1	0	68.3	9.805	113.0	146.8
0	1	1	0	0	1	0	1	0	1	65.1	4.013	108.8	163.0
1	1	1	1	1	0	0	0	0	0	18.8	6.141	23.0	134.9
1	1	1	1	1	1	1	1	1	1	19.7	11.54	27.5	261.4
Average:										42.97	7.874	68.07	176.52



(a)



(b)

Fig. 6. Comparison of delay and power consumption exhibited by the UTBSOI-compressor with optimized value device parameters in (a) rising edge transition, (b) falling edge transition of the input vector.

Table 6. Values of the device parameters used in CMOS and UTBSOI-compressor. (The rest values of the device parameters are kept default).

Parameters	CMOS (gpdk 45 nm)		UTBSOI (BSIM-IMG)	
	n-channel	p-channel	n-channel	p-channel
Channel length (L)	55 nm	55 nm	55 nm	55 nm
Channel length (W)	165 nm	330 nm	165 nm	330 nm
Source diffusion area	45375 nm^2	90750 nm^2	45375 nm^2	90750 nm^2
Drain diffusion area	45375 nm^2	90750 nm^2	45375 nm^2	90750 nm^2
Source diffusion periphery	880 nm	1210 nm	880 nm	1210 nm
Drain diffusion periphery	880 nm	1210 nm	880 nm	1210 nm
Threshold voltage (V_{th})	0.567	−0.544	0.329	−0.365

Table 7. Delay and power consumption exhibited by the CMOS and UTBSOI-compressor at rising and falling edge transition of input vector observed for the combinations given in Table 1.

Rising edge transition:

X_1					X_0					CMOS-compressor		UTBSOI-compressor	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (nW)	Delay (ns)	Power (nW)
0	0	1	1	0	0	1	0	1	0	28.2	0.7275	4.7	1.102
0	1	1	0	0	1	0	1	0	1	26.1	0.7455	4.1	1.059
1	1	1	1	1	0	0	0	0	0	5.4	0.5574	0.5	0.6947
1	1	1	1	1	1	1	1	1	1	6.8	0.8129	0.5	1.024
Average:										16.625	0.7108	2.45	0.9699

Falling edge transition:

X_1					X_0					CMOS-compressor		UTBSOI-compressor	
a_1	b_1	c_1	d_1	e_1	a_0	b_0	c_0	d_0	e_0	Delay (ns)	Power (nW)	Delay (ns)	Power (nW)
0	0	1	1	0	0	1	0	1	0	29.5	0.6366	2.0	0.7993
0	1	1	0	0	1	0	1	0	1	28.9	0.6411	2.0	0.8050
1	1	1	1	1	0	0	0	0	0	5.3	0.5418	0.4	0.6411
1	1	1	1	1	1	1	1	1	1	7.6	0.7831	0.7	0.9141
Average:										17.825	0.65065	1.275	0.78987

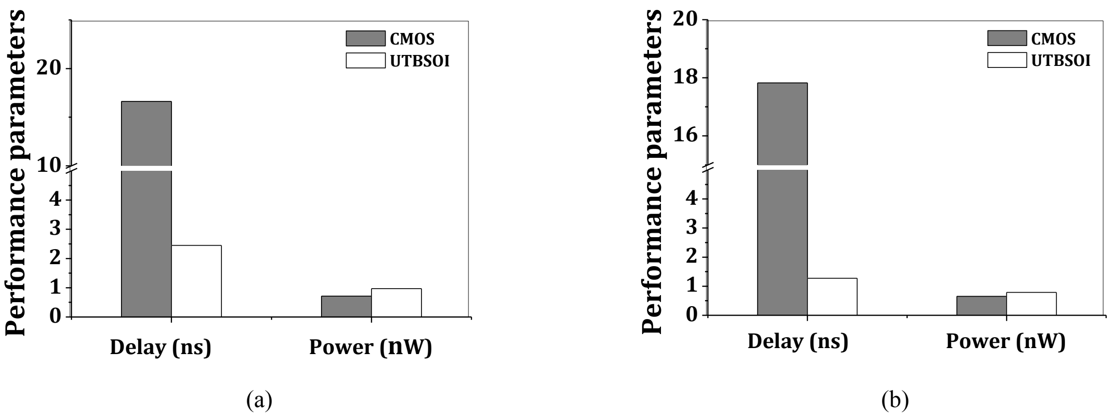


Fig. 7. Comparison of delay and power consumption exhibited by the CMOS and UTBSOI-compressors in the (a) rising edge transition, (b) falling edge transition of input vector.

power consumption at the rising edge and falling edge transition of the input vector. On the other hand, the new device parameter's value has resulted in an increased delay of 75.88% and 58.41% at the rising and falling transition of the input vector [Fig. 6]. Overall, the optimized device parameter values have resulted in a very low PDP = 2.715 fJ of the UTBSOI-compressor.

Advantage of the UTBSOI-compressor has been further clarified from the comparison of its performance parameters with the CMOS counterparts. The CMOS-compressor is simulated using the generic-process design kit (gpdk) 45 nm technology in order to compare its performance parameters with that of UTBSOI-compressor. Table 6 presents the device parameter values of the transistors used in the CMOS and UTBSOI-compressors. Rest of the device parameters of the UTBSOI are kept the same as given in [18]. The performance parameters achieved by the CMOS- as well as UTBSOI- compressors, are given in Table 7. The UTBSOI-compressor has a delay lesser than the CMOS-compressor by 85.26% at rising-edge and 92.84% at the falling-edge transition of the input vector. Power consumption by the CMOS-compressor is less than that of UTBSOI, which can be observed from Fig. 7. The lesser delay in UTBSOI-compressor implies that the use of UTBSOIs in circuits helps in faster applications. Overall, the

UTBSOI-compressor can reduce the PDP by 79.89% than that of CMOS counterparts. Performance comparison of CMOS and UTBSOI-compressors reveal that the UTBSOI can be successfully used in DG-CMOS integrated circuits.

4. Conclusion

In this paper, a DG-CMOS based multi-column 5,5:4 compressor is designed which has been found useful in reducing the number of output bits generated while adding the column of bits (PPA) in Wallace tree multiplier. Reduction of two columns comprising five bits each by the compressor has been simulated through VHDL coding in the device Virtex 6. The UTBSOI transistor has been used to build up the compressor for which the BSIM-IMG model is utilized. The average delays of the UTBSOI-compressor observed from the input pulses of time period 10 µs are 35.3 ns and 68.07 ns at the rising and falling edge transition of the input vector, respectively. The average power consumption of the UTBSOI-compressor at the rising and falling edge transition of the input vector is 76.93 nW and 176.52 nW, respectively. Comparison of performance parameters of the UTBSOI and CMOS-compressor has revealed that the PDP achieved by the UTBSOI-compressor is 79.89% lesser than that of CMOS-compressor. From the simulation results, it can be concluded that the UTBSOI can be successfully used to design any DG-CMOS based circuits from the perspective of delay efficiency (lesser delay).

Acknowledgements

This work is supported by the Ministry of Electronics and Information Technology (MeitY), Government of India, under Visvesvaraya PhD Scheme.

References

[1] H.-S.P. Wong. (2002) "Beyond the conventional transistors." IBM Journal of Research and Development **46 (2/3)**: 133−138.

[2] N. Paydavosi *et al.* (2013) "BSIM-SPICE models enable FinFET and UTB IC designs," IEEE Access **1**: 201−215.

[3] S. Khandelwal *et al.* (2012) "BSIM-IMG: A compact model for ultrathin-body SOI MOSFETs with back-gate control," IEEE Transactions on Electron Devices **59 (8)**: 2019−2026.

[4] M. Cheralathan *et al.* (2013) "Implementation of nanoscale double-gate CMOS circuits using compact advanced transport models," Microelectronics Journal **44(2)**: 80−85.

[5] BSIM-IMG. [Online].(Accessed on 2017, Aug 10). Available: http://bsim.berkeley.edu/models/bsimimg

[6] R. Menon and D. Radhakrishnan.(2006) "High performance 5:2 compressor architecture." In Proc. IEE-Circuits, Devices Syst. **153**: 447−452.

[7] S. Veeramachaneni *et al.* (2007) "Novel architecture for high-speed and low-power 3-2, 4-2, and 5-2 compressors." International Conference on VLSI Design: 324−329.

[8] A. Silveru and M. Bharati. (2013) "Design of compressor based multiplier using degenerate pass transistor logic." International Journal of Engineering Trends and Technology **4(4)**: 896−900.

[9] G.S. Lakshmi, K. Fatima, and B.K. Madhavi. (2010) "Compressor based 8×8 bit vedic multiplier using reversible logic." International Conference on Devices, Circuitsand Systems: 174−178.

[10] A. Dandapat *et al*. (2010) "A 1.2-ns 16×16 bit binary multiplier using high speed compressors." International Journal of Electronics and Communication Engineering : **4(3)**, 485−490.

[11] N. Ravi *et al.* (2010) "Performance evaluation of high speed compressors for high speed compressors for high speed multipliers using 90 nm technology." Recent Advances in Space Technology Services and Climate Change : 189−193.

[12] R. Marimuthu, Y.E. Rezinold, and P.S. Mallick.(2016) "Design and analysis of multiplier using approximate 15-4 compressor." IEEE Access: **5**, 1027−1036.

[13] R.U. Ahmed and P.Saha. (2017) "Design and analysis of new 5,5:4 compressor," International Conference on Computing, Communication, and Sensor Networks: 34−36.

[14] E. Stavinov. (2011) 100 Power Tips for FPGA Designers: Paramount, CA, CreateSpace.

[15] J.P. Uyemura. (2001) CMOS Logic Circuit Design. Dordrecht, Kluwer Academic.

[16] K. Navi *et al.*. (2009) "A novel low-power full-adder cell for low voltage," Integration. VLSI Journal **42(4)**: 457−467.

[17] P. Saha *et al.* (2011) "ASIC design of a high speed low power circuit for factorial calculation using ancient vedic mathematics." Microelectronics Journal 42**(12)**: 1343−1352.

[18] D. Lu *et al.* (2016) "Independent multi-gate MOSFET compact model," Technical Manual, University of California, Berkeley, CA.